

DC

ACORN UNIVERSAL CONSULTANCY

# Director Control

Executive Management Dashboard

|            |                                            |
|------------|--------------------------------------------|
| DOCUMENT   | Full Project Summary & Technical Reference |
| VERSION    | v1.0                                       |
| DATE       | April 2026                                 |
| REPOSITORY | HarshKansara-Acorn / DirectorControl       |
| STACK      | React JS · Node.js / Express · Azure SQL   |

● Phase 1 & 2 Complete — Live on Azure SQL

# DirectorControl — Full Project Summary

---

**Project:** DirectorControl Executive Management Dashboard **Client:** Acorn Universal Consultancy **Repository:** HarshKansara-Acorn / DirectorControl (GitHub) **Document Date:** April 2026 **Current Status:** Phase 1 & 2 Complete — Live on Azure SQL

---

## 1. Project Overview

DirectorControl is a role-based web productivity dashboard built for **Acorn Universal Consultancy**. It serves 3 company directors and their shared Personal Assistant (PA). The PA manages all information centrally from one interface; directors each have a focused, limited-access portal showing only what is relevant to them.

### Business Problem Solved

---

- Directors were receiving updates via email and WhatsApp with no central tracking
- The PA had no structured way to manage reminders, approvals, and tasks across 3 directors simultaneously
- No visibility into upcoming travel, document expiry dates, or bill due dates
- No shared calendar or meeting coordination tool

### Solution Delivered

---

A role-based web portal where:

- **PA (Admin)** manages all 3 directors from one interface with a director switcher dropdown
  - **Directors** each have a focused portal showing only their own tasks, reminders, and approvals
  - All data persists in Azure SQL Server (no mock data in production)
  - Microsoft Teams integration available for calendar and task sync
-

## 2. Tech Stack

| LAYER            | TECHNOLOGY                  | VERSION |
|------------------|-----------------------------|---------|
| Frontend         | React JS                    | 18.3.1  |
| Routing          | React Router DOM            | 6.24.0  |
| Drag & Drop      | @hello-pangea/dnd           | 16.6.0  |
| HTTP Client      | Axios                       | 1.7.2   |
| Date Utilities   | date-fns                    | 3.6.0   |
| Icons            | Lucide React                | 0.395.0 |
| Build Tool       | react-scripts (CRA)         | 5.0.1   |
| Backend          | Node.js + Express           | 4.19.2  |
| Authentication   | JWT (jsonwebtoken)          | 9.0.2   |
| Password Hashing | bcryptjs                    | 2.4.3   |
| Session          | express-session             | 1.18.0  |
| Unique IDs       | uuid                        | 9.0.1   |
| Database Driver  | mssql                       | 11.0.1  |
| Database         | MS SQL Server (Azure)       | —       |
| Query Style      | Raw parameterised SQL       | —       |
| Styling          | CSS Modules + CSS Variables | —       |
| Dev Server       | nodemon                     | 3.1.0   |
| Version Control  | Git + GitHub                | —       |

### Infrastructure

| ITEM              | VALUE                                 |
|-------------------|---------------------------------------|
| Azure SQL Server  | bcdwsqserver.database.windows.net     |
| Database Name     | BCDWsqldatabase                       |
| DB User           | bcsqsvr                               |
| Frontend Dev Port | localhost:3000                        |
| Backend Dev Port  | localhost:5000                        |
| Table Prefix      | DC_ (to avoid conflicts in shared DB) |

### 3. User Roles & Access Matrix

| FEATURE                         | PA (ADMIN)                                          | DIRECTOR                                                     |
|---------------------------------|-----------------------------------------------------|--------------------------------------------------------------|
| Login URL                       | <a href="#">/login</a> → <a href="#">/dashboard</a> | <a href="#">/login</a> → <a href="#">/director/dashboard</a> |
| See all 3 directors             | ✔ Director switcher dropdown                        | ✘ Own data only                                              |
| Create / edit all records       | ✔ Full CRUD                                         | ✘ Read + status updates only                                 |
| Approve / reject requests       | ✘                                                   | ✔                                                            |
| Broadcast to multiple directors | ✔                                                   | ✘                                                            |
| Dashboard tab                   | ✔                                                   | ✔ (director version)                                         |
| Tasks tab (Kanban)              | ✔                                                   | ✔ (own tasks only)                                           |
| Travel tab                      | ✔                                                   | ✘                                                            |
| Documents tab                   | ✔                                                   | ✘                                                            |
| Bills tab                       | ✔                                                   | ✘                                                            |
| Assets tab                      | ✔                                                   | ✘                                                            |
| Events tab                      | ✔                                                   | ✘                                                            |
| Teams tab                       | ✔                                                   | ✘                                                            |
| Reminders tab                   | ✘                                                   | ✔                                                            |
| Approvals tab                   | ✘                                                   | ✔                                                            |
| Profile page                    | ✔                                                   | ✔                                                            |
| Settings page                   | ✔                                                   | ✔                                                            |

#### Login Credentials

| USER       | EMAIL                                       | PASSWORD     |
|------------|---------------------------------------------|--------------|
| PA (Admin) | harsh.kansara@acornuniversalconsultancy.com | Admin@123    |
| Director 1 | director1@acornuniversalconsultancy.com     | Director@123 |
| Director 2 | director2@acornuniversalconsultancy.com     | Director@123 |
| Director 3 | director3@acornuniversalconsultancy.com     | Director@123 |

*Passwords are auto-synced on every backend startup via bcrypt hash update.*

## 4. Feature Inventory

### 4.1 PA Admin Portal ( [/dashboard](#) )

#### Dashboard Tab

- Time-of-day greeting with today's date
- Summary bar: meetings today, unread emails, pending approvals
- 6 widget cards:
  - **Today's Meetings** — time, location, duration; PA can add/delete
  - **Urgent Emails** — unread count, mark-as-read; PA can add/delete
  - **Key Reminders** — priority colour coding (High/Medium/Low); PA can add/delete
  - **Pending Approvals** — Approve/Reject buttons inline; PA can add/delete
  - **Travel Reminders** — upcoming trips with status; PA can add/delete
  - **Task Summary** — overdue / due today / completed counts (auto-calculated)

#### Tasks Tab

- Kanban board with 4 columns: To Do → In Progress → Review → Done
- Drag-and-drop between columns (optimistic UI update + server sync)
- Priority filter: All / High / Medium / Low
- Task cards show: title, description, tags, priority chip, due date
- Add Task modal with director multi-select (broadcast to multiple directors)
- Task detail modal: view full details, update status, delete

#### Travel Tab

- Card grid with status filter: Upcoming / Ongoing / Completed / Cancelled
- Add / Edit / Delete travel records
- Fields: destination, purpose, departure date, return date, notes, status

#### Documents Tab

- Card grid with category filter: Finance / Legal / HR / IT / Admin
- Expiry warning — red highlight when  $\leq 30$  days to expiry
- File type icons: PDF, DOCX, XLSX, PPTX
- Add / Edit / Delete documents
- Fields: title, description, category, file name, file size, file type, expiry date, tags

#### Bills Tab

- Table view with summary stats: pending amount, overdue amount, total count
- Status filter: Pending / Paid / Overdue / Cancelled
- Mark as Paid button with one click
- Multi-currency support: ₹, \$, £, €
- Add / Edit / Delete bills
- Fields: title, vendor, category, amount, currency, due date, invoice number, notes

#### Assets Tab

- Card grid with category + status filters
- Total asset value summary card
- Warranty expiry alerts — red highlight when  $\leq 30$  days
- Toggle maintenance status
- Add / Edit / Delete assets
- Fields: name, description, category, serial number, purchase date, purchase value, current value, location, warranty expiry, assigned to

## Events Tab

- Timeline view grouped by month
- Event type icons: Meeting, Conference, Presentation, Company, Personal
- Priority badges (High / Medium / Low)
- Today's events highlighted in blue
- Shared events visible to multiple directors simultaneously
- Add / Edit / Delete events
- Fields: title, description, type, directors (multi-select), start/end date, start/end time, location, attendees, all-day toggle, priority, notes

## Teams Tab

- Microsoft Teams OAuth2 integration via Azure AD
- Per-director connection status display
- Status bar: presence indicator, today's meetings count, To Do tasks count, OOO status
- Out-of-office banner with message when active
- Calendar tab: all events with filter (All / Teams Meetings / Today)
- Today's Schedule tab: timeline view with Join Meeting buttons
- To Do Tasks tab: grouped by list, overdue items highlighted
- Sync to Events button: imports Teams calendar events into the Events tab
- Setup guide for Azure AD app registration

## Profile Page

- Avatar with colour picker (10 preset colours) + custom initials (max 2 chars)
- First name, last name, email (read-only), phone number
- Job title, department, location / office
- Bio / About Me (500 character limit with counter)
- All data saved to Azure SQL via `/api/users/me/profile`

## Settings Page

- **Password** — change with strength meter + 4 validation rules (length, uppercase, number, special char)
- **Two-Factor Authentication** — enable/disable toggle (saved to DB)
- **Active Sessions** — viewer showing device, IP, last active time
- **Linked Accounts** — Microsoft/Teams, Google, Outlook connect buttons
- **Account Actions** — Sign out everywhere, Deactivate account, Delete account (all require password confirmation)

---

## 4.2 Director Portal ( `/director/*` )

---

### Director Dashboard

- 4 stat cards: Overdue Tasks, Due Today, Pending Tasks, Pending Approvals (all clickable)
- Pending Approvals section with Approve / Reject buttons
- Key Reminders with overdue highlighting
- My Tasks table showing pending tasks with priority, status, due date

### My Tasks Tab

- Full Kanban board (same UX as admin but filtered to own tasks only)
- Drag-and-drop status updates
- Priority filter
- Task detail modal with status update capability

### Reminders Tab

DirectorControl — Project Summary

- All reminders set by the PA for this director

- Filter: All / Active / Overdue
- Days-until-due display
- Overdue items highlighted in red

### Approvals Tab

- Full approval history with tabs: Pending / Approved / Rejected / All
  - One-click Approve / Reject with loading state
  - Approval details: type, from, description, due date, remarks
- 

## 4.3 Cross-Cutting Features

---

### Notification Bell

Aggregates notifications automatically from all modules:

- Pending approvals
- Unread urgent emails
- Overdue bills
- Reminders due within 7 days
- Documents expiring within 30 days
- Asset warranties expiring within 30 days
- Events today or tomorrow
- Teams meetings today (when connected)
- Overdue Teams To Do tasks (when connected)
- Out-of-office active (when connected)

Grouped into Urgent / Important / Info priority tiers. Each notification is clickable and navigates to the relevant tab. Auto-refreshes every 2 minutes.

### Global Search (Ctrl+K)

Searches across all 10 modules simultaneously:

- Tasks, Events, Meetings, Approvals, Reminders, Emails, Travel, Documents, Bills, Assets
- Results grouped by module with match highlighting
- Keyboard navigation: ↑↓ arrows to move, Enter to open, Esc to close
- 220ms debounce to avoid excessive API calls

### Light / Dark / System Theme

- Three-way toggle: Sun (Light) / Moon (Dark) / Monitor (System)
- Implemented via CSS custom properties — instant switch, zero JS overhead
- Persists to localStorage across sessions
- System mode follows OS preference automatically (prefers-color-scheme)

### Broadcast (One-to-Many)

PA can send the same item to multiple directors at once via DirectorSelector component:

- **Tasks** → creates separate independent records per director (each director acts independently)
  - **Reminders** → creates separate records per director
  - **Approvals** → creates separate records per director
  - **Meetings / Events** → creates ONE shared record visible to all selected directors (same event)
-

## 5. Database Schema

All tables use the `DC_` prefix to avoid conflicts in the shared `BCDWSqlDatabase`.

| TABLE                        | PURPOSE                               | KEY COLUMNS                                                                                                                                                 |
|------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>DC_Users</code>        | All users (PA + directors)            | Id, Name, Email, Password, Role, Title, Avatar, FirstName, LastName, Phone, Bio, Location, Department, AvatarColor, TwoFAEnabled, IsActive, LastLoginAt     |
| <code>DC_Tasks</code>        | Tasks assigned to directors           | Id, Title, Description, Priority, Status, AssignedTo, CreatedBy, DueDate, Tags                                                                              |
| <code>DC_Reminders</code>    | Key reminders for directors           | Id, Title, Description, DirectorId, DueDate, Priority, IsActive, CreatedBy                                                                                  |
| <code>DC_Approvals</code>    | Approval requests                     | Id, Type, Title, Description, FromName, FromEmail, DirectorId, Priority, Status, Remarks, ActionBy, ActionAt                                                |
| <code>DC_Meetings</code>     | Meetings (individual + shared)        | Id, Title, DirectorId, DirectorIds, IsShared, MeetingDate, MeetingTime, Duration, Location, Attendees                                                       |
| <code>DC_UrgentEmails</code> | Urgent email alerts                   | Id, Subject, FromEmail, FromName, DirectorId, Preview, Priority, IsRead                                                                                     |
| <code>DC_Travel</code>       | Travel records                        | Id, Destination, Purpose, DirectorId, DepartureDate, ReturnDate, Status, Notes                                                                              |
| <code>DC_Documents</code>    | Document registry                     | Id, Title, Category, DirectorId, FileName, FileSize, FileType, ExpiryDate, Tags                                                                             |
| <code>DC_Bills</code>        | Bills and invoices                    | Id, Title, Vendor, Category, DirectorId, Amount, Currency, DueDate, Status, InvoiceNumber, PaidDate                                                         |
| <code>DC_Assets</code>       | Asset registry                        | Id, Name, Category, DirectorId, SerialNumber, PurchaseDate, PurchaseValue, CurrentValue, Status, WarrantyExpiry, AssignedTo                                 |
| <code>DC_Events</code>       | Calendar events (individual + shared) | Id, Title, Type, DirectorId, DirectorIds, IsShared, StartDate, EndDate, StartTime, EndTime, Location, Attendees, Priority, Status, TeamsId, JoinUrl, Source |
| <code>DC_TeamsTokens</code>  | OAuth tokens for Teams                | DirectorId, AccessToken, RefreshToken, ExpiresAt, MsUserEmail                                                                                               |

### Schema Management

- Schema runs automatically on every backend startup via `setupDb.js`
- All `CREATE TABLE` statements use `IF NOT EXISTS` guards — safe to restart without data loss
- Profile columns (FirstName, LastName, Phone, Bio, etc.) added via `migrateProfile.js` migration
- Shared event columns (DirectorIds, IsShared) added via the same migration



## 6. API Endpoints

### Authentication

| METHOD | ENDPOINT                      | AUTH | DESCRIPTION                            |
|--------|-------------------------------|------|----------------------------------------|
| POST   | <code>/api/auth/login</code>  | None | Login, returns JWT token + user object |
| POST   | <code>/api/auth/logout</code> | JWT  | Logout                                 |

### Users / Profile

| METHOD | ENDPOINT                            | AUTH  | DESCRIPTION                                 |
|--------|-------------------------------------|-------|---------------------------------------------|
| GET    | <code>/api/users/directors</code>   | Admin | List all directors                          |
| GET    | <code>/api/users/me</code>          | JWT   | Get current user full profile               |
| PUT    | <code>/api/users/me/profile</code>  | JWT   | Update profile (name, phone, bio, etc.)     |
| PUT    | <code>/api/users/me/avatar</code>   | JWT   | Update avatar colour and initials           |
| PUT    | <code>/api/users/me/password</code> | JWT   | Change password (requires current password) |
| PUT    | <code>/api/users/me/2fa</code>      | JWT   | Toggle 2FA on/off                           |
| DELETE | <code>/api/users/me</code>          | JWT   | Deactivate or permanently delete account    |
| GET    | <code>/api/users/me/sessions</code> | JWT   | List active sessions                        |

### Tasks

| METHOD | ENDPOINT                            | AUTH  | DESCRIPTION                        |
|--------|-------------------------------------|-------|------------------------------------|
| GET    | <code>/api/tasks?directorId=</code> | JWT   | List tasks for a director          |
| POST   | <code>/api/tasks</code>             | Admin | Create task for single director    |
| POST   | <code>/api/tasks/broadcast</code>   | Admin | Create task for multiple directors |
| PUT    | <code>/api/tasks/:id</code>         | Admin | Update task                        |
| PATCH  | <code>/api/tasks/:id/status</code>  | JWT   | Update task status only            |
| DELETE | <code>/api/tasks/:id</code>         | Admin | Delete task                        |

## Reminders

| METHOD | ENDPOINT                                | AUTH  | DESCRIPTION                            |
|--------|-----------------------------------------|-------|----------------------------------------|
| GET    | <code>/api/reminders?directorId=</code> | JWT   | List reminders                         |
| POST   | <code>/api/reminders</code>             | Admin | Create reminder                        |
| POST   | <code>/api/reminders/broadcast</code>   | Admin | Create reminder for multiple directors |
| PUT    | <code>/api/reminders/:id</code>         | Admin | Update reminder                        |
| DELETE | <code>/api/reminders/:id</code>         | Admin | Delete reminder                        |

## Approvals

| METHOD | ENDPOINT                                | AUTH     | DESCRIPTION                            |
|--------|-----------------------------------------|----------|----------------------------------------|
| GET    | <code>/api/approvals?directorId=</code> | JWT      | List approvals                         |
| POST   | <code>/api/approvals</code>             | Admin    | Create approval request                |
| POST   | <code>/api/approvals/broadcast</code>   | Admin    | Create approval for multiple directors |
| PATCH  | <code>/api/approvals/:id/action</code>  | Director | Approve or reject                      |
| DELETE | <code>/api/approvals/:id</code>         | Admin    | Delete approval                        |

## Meetings

| METHOD | ENDPOINT                                         | AUTH  | DESCRIPTION                       |
|--------|--------------------------------------------------|-------|-----------------------------------|
| GET    | <code>/api/meetings?directorId=&amp;date=</code> | JWT   | List meetings (includes shared)   |
| POST   | <code>/api/meetings</code>                       | Admin | Create meeting (single or shared) |
| PUT    | <code>/api/meetings/:id</code>                   | Admin | Update meeting                    |
| DELETE | <code>/api/meetings/:id</code>                   | Admin | Delete meeting                    |

## Other Modules (same CRUD pattern)

| MODULE        | BASE PATH                   | EXTRA ENDPOINTS                |
|---------------|-----------------------------|--------------------------------|
| Urgent Emails | <code>/api/emails</code>    | PATCH <code>/:id/read</code>   |
| Travel        | <code>/api/travel</code>    | —                              |
| Documents     | <code>/api/documents</code> | —                              |
| Bills         | <code>/api/bills</code>     | PATCH <code>/:id/status</code> |
| Assets        | <code>/api/assets</code>    | PATCH <code>/:id/status</code> |
| Events        | <code>/api/events</code>    | PATCH <code>/:id/status</code> |

## Teams Integration

| METHOD | ENDPOINT                                               | DESCRIPTION                                 |
|--------|--------------------------------------------------------|---------------------------------------------|
| GET    | <code>/api/teams/auth/connect?directorId=</code>       | Start OAuth2 flow                           |
| GET    | <code>/api/teams/auth/callback</code>                  | OAuth2 callback handler                     |
| POST   | <code>/api/teams/auth/disconnect</code>                | Disconnect Teams for a director             |
| GET    | <code>/api/teams/status?directorId=</code>             | Connection status                           |
| GET    | <code>/api/teams/summary?directorId=</code>            | Full Teams data (presence, calendar, tasks) |
| GET    | <code>/api/teams/calendar?directorId=&amp;days=</code> | Calendar events                             |
| GET    | <code>/api/teams/today?directorId=</code>              | Today's schedule                            |
| GET    | <code>/api/teams/tasks?directorId=</code>              | To Do tasks                                 |
| GET    | <code>/api/teams/presence?directorId=</code>           | User presence status                        |
| POST   | <code>/api/teams/sync</code>                           | Sync Teams calendar into Events tab         |

## System / Admin

| METHOD | ENDPOINT                                    | DESCRIPTION                         |
|--------|---------------------------------------------|-------------------------------------|
| GET    | <code>/api/health</code>                    | Health check + DB connection status |
| POST   | <code>/api/db/setup</code>                  | Manually trigger schema setup       |
| POST   | <code>/api/admin/reset-all-directors</code> | Reset all director passwords        |
| POST   | <code>/api/migrate/reset-passwords</code>   | One-time password migration         |

## 7. Project File Structure

```
DirectorControl/
├── backend/
│   ├── schema.sql           # DB schema – runs on startup
│   ├── package.json         # Node dependencies
│   ├── .env                 # Azure SQL + JWT secrets
│   ├── .env.example         # Template for new devs
│   └── src/
│       ├── index.js         # Express server entry point
│       ├── config/
│       │   ├── db.js        # Azure SQL connection pool
│       │   ├── setupDb.js    # Auto-run schema on startup
│       │   └── migrateProfile.js # Add profile + shared-event columns
│       ├── middleware/
│       │   └── auth.js       # JWT verification middleware
│       ├── data/
│       │   └── mockData.js    # Legacy mock data (preserved, not used)
│       ├── services/
│       │   └── teamsService.js # Microsoft Graph API service layer
│       └── routes/
│           ├── auth.js       # Login / logout
│           ├── users.js      # Profile + settings endpoints
│           ├── tasks.js      # Tasks + broadcast
│           ├── reminders.js   # Reminders + broadcast
│           ├── approvals.js   # Approvals + broadcast + action
│           ├── meetings.js    # Meetings + shared meetings
│           ├── emails.js      # Urgent emails
│           ├── travel.js      # Travel records
│           ├── documents.js   # Document registry
│           ├── bills.js       # Bills + mark paid
│           ├── assets.js      # Assets + maintenance toggle
│           ├── events.js      # Events + shared events
│           ├── teams.js       # Teams OAuth + data endpoints
│           ├── admin.js       # Password reset utilities
│           └── migrate.js     # One-time migration endpoints
│
├── frontend/
│   ├── package.json         # React dependencies
│   ├── .env                 # REACT_APP_API_URL
│   ├── .env.example
│   └── src/
│       ├── App.js           # Route definitions (admin + director)
│       ├── index.css        # CSS custom properties (light + dark)
│       ├── services/
│       │   └── api.js        # Axios instance with base URL + auth header
│       ├── context/
│       │   ├── AuthContext.js # Login, logout, updateUser
│       │   ├── DirectorContext.js # Director switcher state
│       │   ├── NotificationContext.js # Aggregated notifications (auto-refresh)
│       │   ├── SearchContext.js # Global search state
│       │   └── ThemeContext.js # Light / Dark / System theme
│       ├── components/
│       │   ├── auth/
│       │   │   └── ProtectedRoute.js # Role-based route guard
│       │   ├── layout/
│       │   │   ├── DashboardLayout.js # PA admin shell (header + nav)
│       │   │   ├── DirectorLayout.js # Director shell (header + nav)
│       │   │   └── NotificationPanel.js # Bell dropdown panel
```

```

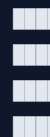
| | | | └─ SearchPanel.js      # Ctrl+K search overlay
| | | | └─ ThemeToggle.js     # Sun/Moon/System toggle button
| | | └─ dashboard/
| | |   └─ DashboardCard.js    # Base card wrapper
| | |   └─ MeetingsCard.js
| | |   └─ UrgentEmailsCard.js
| | |   └─ RemindersCard.js
| | |   └─ ApprovalsCard.js
| | |   └─ TravelCard.js
| | |   └─ TaskSummaryCard.js
| | └─ modals/
| |   └─ Modal.js              # Base modal wrapper
| |   └─ FormField.js          # Reusable form field
| |   └─ DirectorSelector.js   # Multi-director broadcast selector
| |   └─ AddTaskModal.js
| |   └─ AddReminderModal.js
| |   └─ AddApprovalModal.js
| |   └─ AddMeetingModal.js
| |   └─ AddEmailModal.js
| |   └─ AddTravelModal.js
| |   └─ TaskDetailModal.js
| └─ pages/
|   └─ LoginPage.js
|   └─ Dashboard.js
|   └─ Tasks.js
|   └─ Travel.js
|   └─ Documents.js
|   └─ Bills.js
|   └─ Assets.js
|   └─ Events.js
|   └─ Teams.js
|   └─ Profile.js
|   └─ Settings.js
|   └─ director/
|     └─ DirectorDashboard.js
|     └─ DirectorTasks.js
|     └─ DirectorReminders.js
|     └─ DirectorApprovals.js
|
└─ scripts/
  └─ verify-and-commit.sh      # Auto-commit helper script
└─ .kiro/                      # Kiro IDE config + hooks
└─ .gitignore
└─ package.json                # Root workspace package
└─ README.md

```

## 8. Development Gantt Chart

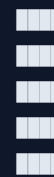


DirectorSelector multi-select component  
Broadcast tasks / reminders / approvals  
Shared meetings (one record, multiple directors)  
Shared events (one record, multiple directors)



#### SPRINT 9 – Polish & Fixes

Profile email overflow fix (word-break CSS)  
CSS variable theming across all pages  
Proxy removal (hot-reload fix)  
Auto-commit Kiro hooks setup  
Project summary document



TOTAL DURATION

|————— ~9 Sprints / ~9 Weeks —————|

## 9. Key Design Decisions

| DECISION                               | RATIONALE                                                                                                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CSS Modules over Tailwind              | Scoped styles per component, no class name conflicts, easier to maintain                                                                                                           |
| CSS Custom Properties for theming      | Zero JS overhead for dark mode, instant switch, no re-render                                                                                                                       |
| Raw parameterised SQL over ORM         | Direct control, no abstraction overhead, easier to debug Azure SQL issues                                                                                                          |
| Mock data preserved but unused         | <code>mockData.js</code> kept for reference/offline testing; all routes use real DB                                                                                                |
| Broadcast vs Shared records            | Tasks/reminders/approvals are broadcast (independent copies) so each director acts independently; meetings/events are shared (one record) since they represent the same occurrence |
| Role-based URL namespacing             | <code>/dashboard</code> for PA, <code>/director/*</code> for directors — clean separation, no accidental cross-access                                                              |
| Auto schema on startup                 | <code>IF NOT EXISTS</code> guards make it safe to run on every restart without manual intervention                                                                                 |
| JWT in localStorage                    | Appropriate for this internal intranet tool; for public-facing production consider httpOnly cookies                                                                                |
| Director passwords separate from admin | <code>Director@123</code> vs <code>Admin@123</code> avoids Chrome data breach warnings for shared passwords                                                                        |

## 10. Running the Project

### Prerequisites

- Node.js 18+
- Git Bash (Windows) — required for git commands
- Network access to `bcdwsqserver.database.windows.net`

## Start Backend

---

```
cd backend
npm install
npm run dev
```

Expected startup output:

```
🚀 Server running on port 5000
✅ Connected to Azure MS SQL Server
✅ Schema ready – 12 statements ran
✅ Profile & shared-event columns ready
✅ Passwords synced – PA: Admin@123 | Directors: Director@123
```

## Start Frontend

---

```
cd frontend
npm install
npm start
```

Opens at <http://localhost:3000>

## Health Check

---

```
GET http://localhost:5000/api/health
→ { "status": "OK", "db": "connected", "message": "DirectorControl API running" }
```

## Manual Schema Reset (if needed)

---

```
POST http://localhost:5000/api/db/setup
```



## 11. Pending / Future Work (Phase 3+)

| FEATURE                               | PRIORITY | NOTES                                                              |
|---------------------------------------|----------|--------------------------------------------------------------------|
| Real director names in DB             | High     | Update <code>DC_Users</code> with actual director names from Acorn |
| Azure AD app registration for Teams   | High     | Needs client ID + secret from Azure portal to enable OAuth         |
| File upload for Documents tab         | Medium   | Currently stores metadata only — no actual file binary             |
| Email notifications                   | Medium   | Notify directors when PA adds approvals or reminders               |
| Push notifications                    | Medium   | Browser push API for urgent items                                  |
| Audit log                             | Medium   | Track who changed what and when                                    |
| Mobile responsive improvements        | Low      | Currently desktop-first layout                                     |
| Export to PDF / Excel                 | Low      | For bills, assets, tasks reports                                   |
| Recurring tasks / reminders           | Low      | Weekly / monthly repeat options                                    |
| Calendar grid view for Events         | Low      | Month / week grid in addition to timeline                          |
| Two-Factor Authentication (full TOTP) | Low      | Currently toggle only — no actual authenticator app flow           |
| Director name customisation           | Low      | Allow PA to rename directors in settings                           |

Document generated: April 2026 / DirectorControl v1.0 / Acorn Universal Consultancy